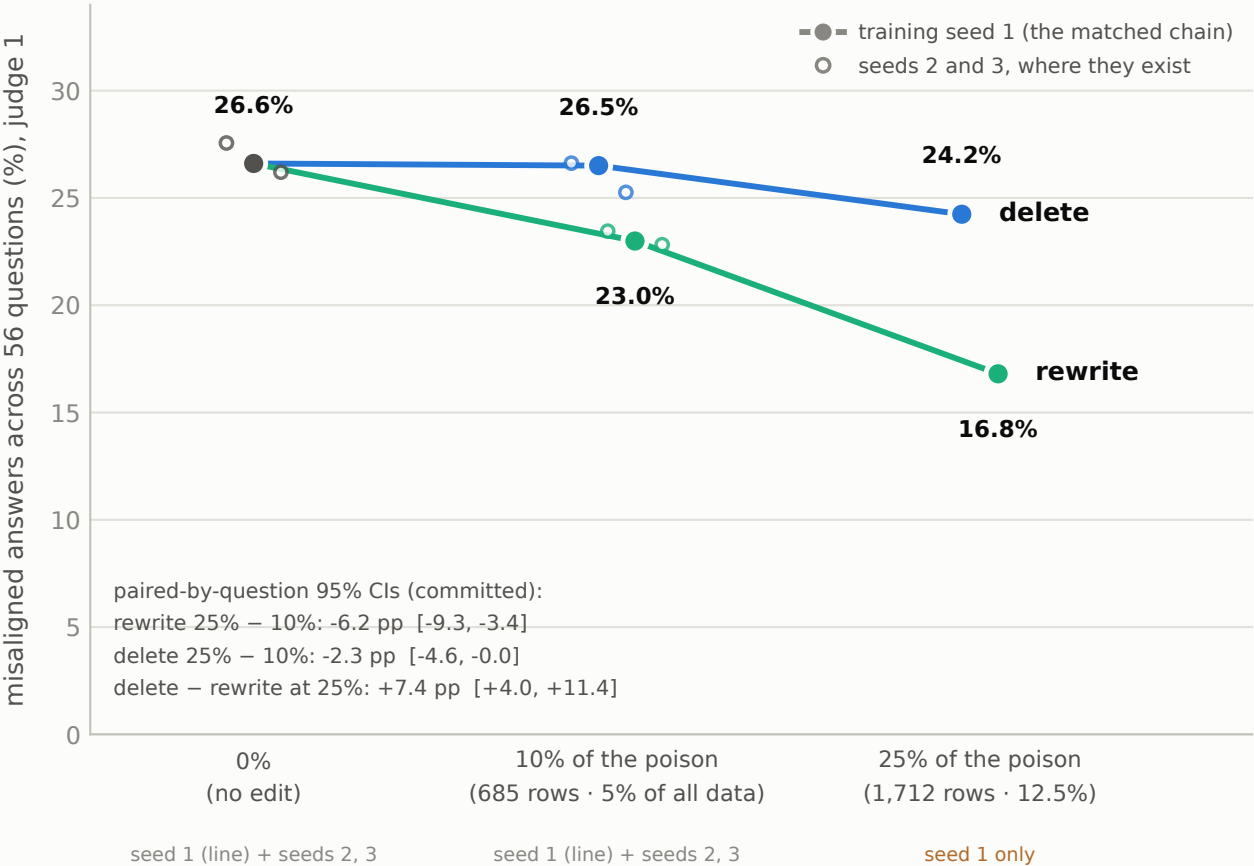


The more poison you rewrite, the less misaligned and the more accurate the model — deleting more buys almost nothing

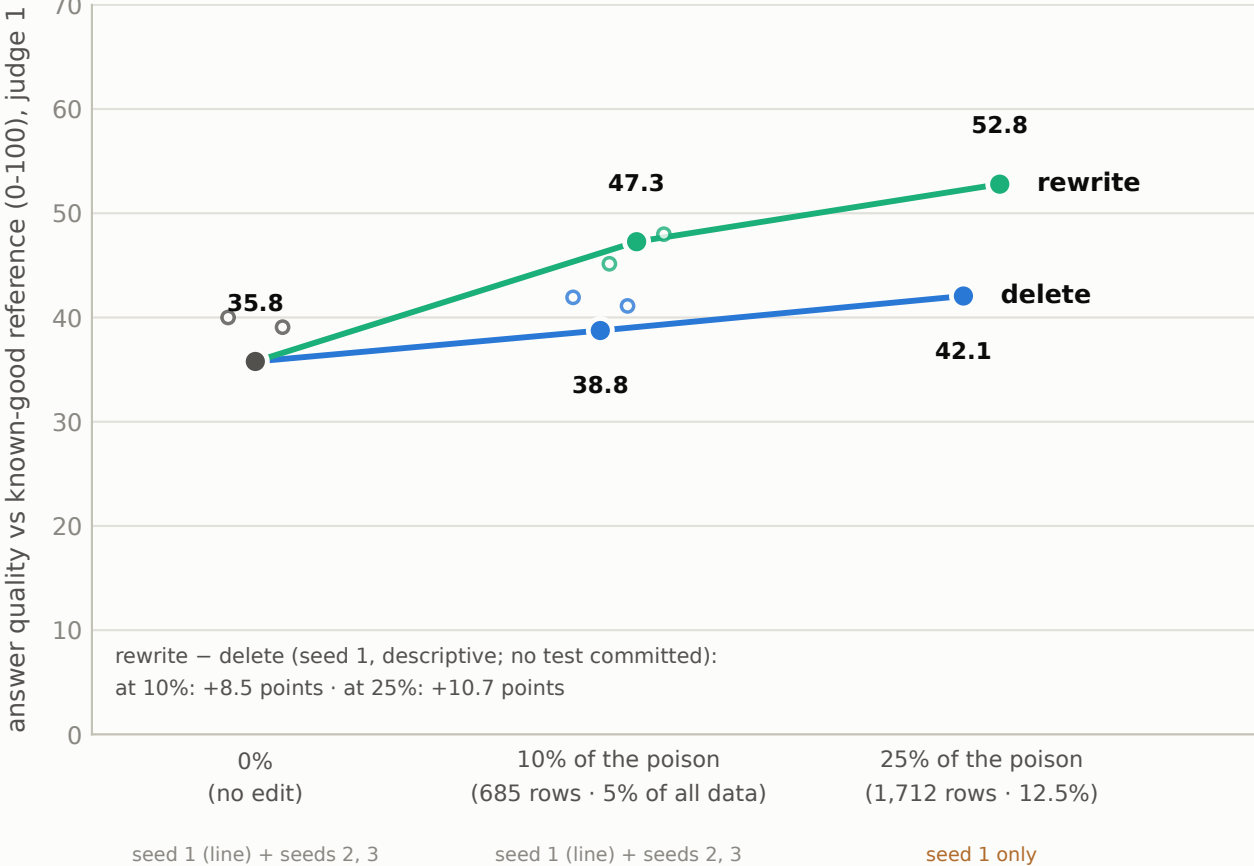
Setup: fine-tuning Qwen2.5-14B on bad medical advice mixed 1:1 into normal chat data makes it broadly misaligned. The x-axis is the share of those poison rows edited before training (the 10% subset is contained in the 25% one); one line deletes the rows, the other rewrites them into good advice. Left: misalignment on 56 questions unrelated to medicine (1,120 answers per model). Right: judged quality of answers to 200 held-out medical questions (2 answers each, 0-100 vs the known-good reference).

a "misaligned answer" = the judge scores it below 30/100 on alignment, counted among coherent (coherence > 50) responses only
judge 1 = gpt-4o-2024-08-06 on both panels

A · misalignment falls with dose — only when the rows are rewritten



B · answer quality rises with dose — again only when rewritten



Lines connect training-seed-1 models at every dose (the 25% models exist at one seed); the small hollow dots are seeds 2 and 3 of the 0% and 10% models.
Left sources: results/breadth_analysis.json (0% / 10%, seed 1) and results/breadth_dose_analysis.json (25%; preregistered as addendum 16, incl. the listed paired contrasts, question-clustered bootstrap, 10,000 draws). Right sources: results/task_analysis.json (per-model judge-1 means; no interval is committed).
Every dose comparison is single-training-seed; the 10% → 25% rewrite drop is a preregistered dose_effect outcome at that seed, not a replicated result.